
ProbSol Platform – Technical Report

1. Introduction

ProbSol is a full-stack, credit-based problem-solving platform designed to connect students with industry professionals. The system enables industry users to post real-world challenges, while students can submit solutions and earn credits based on evaluation outcomes.

The platform promotes collaborative learning, practical problem-solving, and skill validation in a structured digital environment.

2. Technology Stack

2.1 Backend Technologies

The backend is built using a modern JavaScript runtime and follows RESTful API principles:

- **Node.js** – Server-side runtime environment
- **Express.js** – Web application framework for routing and middleware
- **MongoDB** – NoSQL database for flexible data storage
- **Mongoose** – ODM for schema modeling and database interaction

Security & Authentication

- **bcryptjs** – Secure password hashing
- **jsonwebtoken (JWT)** – Token-based authentication and authorization

Utilities

- **CORS** – Enables cross-origin communication between frontend and backend
 - **dotenv** – Manages environment variables securely
 - **Nodemon** – Development tool for automatic server restarts
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2.2 Frontend Technologies

The frontend is developed as a responsive Single Page Application (SPA):

- **React 18** – Component-based UI library with hooks
- **React Router v6** – Client-side routing and navigation
- **Axios** – HTTP client for API communication

Styling & Build Tools

- **Tailwind CSS** – Utility-first CSS framework for rapid UI design
 - **PostCSS & Autoprefixer** – CSS processing and browser compatibility
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3. System Architecture

ProbSol follows a **client-server architecture**:

- **Frontend (React)** handles UI/UX and user interactions
- **Backend (Node.js/Express)** manages business logic and APIs
- **Database (MongoDB)** stores persistent application data

Architecture Flow

1. User interacts with frontend
 2. API requests sent via Axios
 3. Backend processes requests and validates JWT
 4. Database queried/updated
 5. Response returned to frontend
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4. User Roles

The platform supports two primary user roles:

4.1 Students

- Explore available problems
- Submit solutions
- Track submission history
- Earn credits based on performance

4.2 Industry Users

- Post real-world problems
 - Review student submissions
 - Assign grades and credits
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5. Key Features

Feature	Description
Dashboard	Overview of platform activity and user stats
Explore Problems	Browse available challenges
Problem Details	View descriptions, deadlines, and rewards
Submit Solution	Upload solutions for evaluation
My Submissions	Track past submissions and results
Post Problem	Industry users create challenges
Review Submissions	Evaluate and grade student work
Profile Management	View user details and earned credits

6. Data Flow

The system follows a structured workflow:

- 1. Authentication**
 - Users log in/register
 - JWT token is generated and stored
 - 2. Problem Interaction**
 - Students browse and select problems
 - Industry users post new challenges
 - 3. Submission Process**
 - Students submit solutions
 - Submissions stored in database
 - 4. Evaluation**
 - Industry users review submissions
 - Credits assigned based on performance
 - 5. Credit System**
 - Credits updated in user profile
 - Acts as a reward and reputation metric
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7. Database Schema Design

7.1 Users Collection

- `name` – User's full name
- `email` – Unique identifier

- **password** – Hashed password
 - **role** – Either *student* or *industry*
 - **credits** – Accumulated reward points
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7.2 Problems Collection

- **title** – Problem name
 - **description** – Detailed explanation
 - **domain** – Category (e.g., AI, Web Dev)
 - **difficulty** – Easy/Medium/Hard
 - **creditReward** – Points awarded
 - **deadline** – Submission cutoff date
 - **createdBy** – Reference to industry user
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7.3 Submissions Collection

- **studentId** – Reference to student
 - **problemId** – Associated problem
 - **solution** – Submitted answer/content
 - **status** – Pending/Reviewed
 - **grade** – Evaluation result
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8. Security Considerations

- Passwords are securely hashed using **bcrypt**
 - Authentication handled via **JWT tokens**
 - Protected routes ensure role-based access control
 - Environment variables stored securely using **dotenv**
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9. Advantages of the System

- Encourages **practical learning**
- Bridges gap between **academia and industry**
- Scalable due to **NoSQL database design**
- Modular and maintainable architecture
- Real-time credit-based motivation system

10. Conclusion

ProbSol is a robust, scalable, and efficient platform that facilitates real-world problem-solving collaboration. By integrating modern web technologies with a structured credit-based system, it provides meaningful engagement for both students and industry professionals.

The architecture ensures flexibility, security, and performance, making it suitable for future enhancements such as AI-based grading, real-time collaboration, and analytics dashboards.
